

MODULE 5

# **EARTH AND ENVIRONMENTAL SCIENCE**

## 5.1 Introduction to Climate Science: Why the Climate Matters

Climate science is the branch of Earth science that focuses on understanding the climate system and the way it changes over long periods of time. Climate is defined as the long-term average pattern of weather variables such as temperature, rainfall, humidity and wind, usually calculated over a period of at least thirty years. Unlike weather, which may vary from day to day, climate describes the broad trends and seasonal patterns that influence ecosystems, agriculture, water resources and human livelihoods.

The Earth's climate system is composed of several interacting components: the atmosphere, oceans, land surface, snow and ice (the cryosphere) and the biosphere. These components exchange heat, moisture and energy continuously. Because the system is highly interconnected, even small changes in one component can lead to significant feedback effects across the globe. For example, melting ice reduces reflection of sunlight, leading to additional warming.

Climate matters because it sustains life on Earth. Stable climate conditions support food production, water supply, biodiversity and economic development. However, increasing greenhouse gas emissions have led to rapid climate change. Rising temperatures increase the likelihood and severity of floods, droughts, tropical storms and heatwaves. These events affect agriculture, infrastructure, human health and ecosystems, and contribute to social and economic instability.

Natural drivers of climate variability include changes in solar radiation, volcanic eruptions and variations in ocean circulation. In recent decades, however, human activities such as burning fossil fuels, deforestation and industrial emissions have become the dominant cause of climate change. Carbon dioxide, methane and nitrous oxide trap heat in the atmosphere, strengthening the greenhouse effect. As a result, global temperatures, sea levels and extreme weather events continue to rise.

## 5.2 Climate Models and Extreme Weather

Climate models are mathematical tools that simulate the behaviour of the climate system based on physical laws such as thermodynamics and fluid dynamics. These models divide the Earth into a three-dimensional grid and compute changes over time. Simple models only examine global energy balance, while advanced Earth System Models also include vegetation growth, chemical reactions and human influences.

Extreme weather events are becoming more frequent and intense as the world warms. Floods occur more often because warmer air can hold more moisture, leading to

heavier rainfall. Sea-level rise also increases the risk of coastal flooding. Conversely, droughts occur when rainfall decreases or becomes irregular, and higher temperatures increase evaporation. Heatwaves have become longer and more intense, especially in urban areas where concrete and asphalt trap heat. These changes result in loss of life, disruption of livelihoods, agricultural decline and greater pressure on healthcare and emergency services.

## 5.3 Environmental Pollutants

Environmental pollutants are substances or forms of energy that enter the environment and cause harmful effects on ecosystems, human health and infrastructure. Pollutants may exist in gaseous, liquid or solid form, and may be introduced into the atmosphere, water bodies or soil. They can originate from natural sources such as volcanic eruptions, but in the modern world most pollution arises from human activities including industry, transportation, agriculture and urbanisation.

Pollutants can be classified in several ways. Air pollutants include carbon monoxide, sulphur dioxide and particulate matter. Water pollutants include industrial effluents, sewage and agricultural run-off. Soil pollutants include pesticides and heavy metals. Some pollutants are biodegradable and break down naturally over time, while others are non-biodegradable and persist in the environment for decades, accumulating in living organisms and food chains.

## 5.4 Persistent Organic Pollutants (POPs)

Persistent Organic Pollutants, commonly referred to as POPs, are a group of toxic chemical compounds that are resistant to environmental degradation. Because of their persistence, they remain in soil, water and air for many years and are transported long distances by wind and ocean currents. POPs are fat-soluble and tend to accumulate in the tissues of animals and humans, becoming more concentrated at higher levels of the food chain, a process known as biomagnification.

Important examples of POPs include the pesticide DDT, polychlorinated biphenyls (PCBs) used in electrical equipment, and dioxins formed during waste incineration and industrial chemical processes. Many of these substances were widely used because they were chemically stable, effective and inexpensive. However, their environmental and health impacts have proven extremely severe.

Sources of POP release include pesticide application, leakage from industrial materials, chemical manufacturing, accidental spills and incomplete waste combustion. In developing countries, obsolete pesticide stockpiles are a major concern.

The harmful effects of POPs are wide-ranging. They interfere with hormonal systems, affect fertility, damage the immune and nervous systems and increase the risk of cancer. Prenatal and early-life exposure may cause developmental disorders, learning disabilities and birth defects. Wildlife including birds, marine mammals and fish are also affected. Due to these concerns, the Stockholm Convention (2001) was introduced to eliminate or restrict POPs globally.

## 5.5 Solid Waste Management

Solid waste refers to discarded materials from households, businesses, industries, hospitals and institutions. With increasing population, urbanisation and consumerism, waste generation has increased dramatically. If waste is not managed properly, it leads to environmental pollution, spread of disease, foul odour, visual pollution and land degradation.

Solid Waste Management (SWM) refers to the systematic handling of waste from the point of generation to its final disposal. The process consists of several interconnected stages.

Waste is first generated at households, markets, industries and institutions. It is then stored in containers or bins. Proper storage prevents littering and reduces the spread of insects and rodents. Waste is collected either door-to-door or from community bins by municipal authorities. It is then transported to transfer stations or treatment facilities.

Segregation is a critical step, where waste is separated into biodegradable waste, recyclable materials and hazardous waste. Biodegradable waste such as kitchen scraps can be treated through composting or anaerobic digestion to produce manure and biogas. Recyclable materials such as paper, metals and plastics are processed and converted into new products, saving energy and raw materials. Some waste is incinerated to reduce volume and generate energy, although emissions must be controlled. Finally, non-recyclable and inert waste is disposed of in sanitary landfills designed to prevent leachate contamination.

The 3R principle — Reduce, Reuse and Recycle — plays a key role in sustainable waste management. Reducing consumption lowers total waste generated. Reuse extends product life. Recycling minimises resource extraction. Examples include composting food waste, using cloth bags, repairing appliances and recycling packaging materials.

## 5.6 Heavy Metal Pollution: Sources and Health Effects

Heavy metals such as lead, mercury and arsenic are highly toxic even at very low concentrations. Unlike organic pollutants, these metals do not degrade or disappear. Instead, they accumulate in soil, water, plants and living organisms.

Lead has historically been used in paints, plumbing pipes, batteries and fuels. People may be exposed through contaminated water, workplace exposure or household dust. Lead poisoning damages the nervous system, reduces intelligence in children, causes behavioural problems and leads to kidney and cardiovascular disease in adults.

Mercury is released from coal-fired power plants, industrial processes and artisanal gold mining. In aquatic environments it forms methylmercury, which accumulates in fish. Consumption of contaminated seafood leads to neurological damage, tremors, memory loss and developmental disorders in unborn children. The Minamata tragedy in Japan is a well-known example.

Arsenic occurs naturally in groundwater in some regions and is also released from mining, fertilizers and burning coal. Long-term exposure leads to skin lesions, cancers of the lung and bladder, cardiovascular disorders and neurological impairment. Millions of people in Bangladesh and India continue to face arsenic contamination in drinking water.

Because heavy metals persist in the environment and bioaccumulate, they pose long-term risks to ecosystems and human populations.

## 5.7 Electronic Waste (E-waste): Sources, Composition and Recycling

Electronic waste, commonly called e-waste, refers to discarded electrical and electronic devices such as computers, mobile phones, televisions, printers and household appliances. Rapid technological development, falling prices and consumer demand for the latest devices have caused e-waste to become one of the fastest growing waste streams in the world.

The composition of e-waste is complex. It contains plastics, glass, circuit boards, precious metals such as gold and silver, base metals including copper and aluminium, and hazardous metals such as lead, mercury and cadmium. Plastics often contain brominated flame retardants, which release highly toxic emissions when burned.

When e-waste is disposed of improperly in landfills or informal recycling sites, haz-

ardous substances leach into soil and groundwater. Burning e-waste releases toxic smoke and particulate matter, leading to respiratory diseases and environmental contamination. Workers in informal recycling, often without protective equipment, are exposed to chemical burns, metal poisoning and cancer-causing fumes. Children and pregnant women are especially vulnerable.

However, e-waste also contains valuable materials that can be recovered through proper recycling. Formal recycling facilities dismantle devices safely, separate components mechanically, and extract precious and base metals using controlled processes. This reduces environmental impact and conserves natural resources.

The circular economy approach aims to redesign products for longer life, repairability and recyclability. Policies such as Extended Producer Responsibility require manufacturers to manage end-of-life collection and recycling.

## 5.8 Groundwater Depletion: Causes and Consequences

Groundwater stored in aquifers beneath the Earth's surface is one of the most important freshwater sources, particularly in regions where rainfall is seasonal or surface water is scarce. Over-extraction occurs when groundwater is pumped out faster than it is naturally recharged by rainfall infiltration.

Agriculture is the largest consumer of groundwater, especially in regions cultivating water-intensive crops such as rice and sugarcane. Urbanisation and industrialisation further increase demand. Unregulated borewells and subsidised electricity for pumping often encourage excessive withdrawal. Climate change, by altering rainfall patterns and increasing drought frequency, reduces recharge and increases dependence on groundwater.

The consequences of groundwater over-extraction are severe. Water tables fall, forcing wells to be drilled deeper, increasing pumping costs. Poor households may lose access to drinking water entirely. Land subsidence occurs when aquifers collapse, damaging buildings, roads and pipelines. In coastal regions, seawater intrusion contaminates freshwater aquifers with salt. Rivers and wetlands dependent on groundwater inflow may dry up, harming ecosystems.

Sustainable groundwater management requires regulating extraction, promoting efficient irrigation methods such as drip irrigation, protecting recharge zones and encouraging rainwater harvesting and artificial recharge structures. Public education and policy enforcement are essential to prevent irreversible damage.

## 5.9 Conclusion

Environmental issues such as climate change, pollution, waste mismanagement, toxic chemicals, e-waste and groundwater depletion are deeply interconnected. They influence ecosystems, human health, economic stability and social equity. Scientific understanding, responsible policy, technological innovation and public awareness together provide the pathway towards sustainable environmental management. Students and future professionals play a vital role in developing and implementing solutions for a healthier planet.

## 5.10 Model Questions

### Long Answer / Essay Type Questions

1. Discuss how certain synthetic organic chemicals remain in the environment for long periods and accumulate in living organisms. Explain their sources and the risks they pose to human health and ecosystems.
2. Explain the complete pathway followed by solid waste from the point of generation to its final disposal. Use suitable real-life examples to highlight the importance of each stage in the process.
3. Industrialisation and urbanisation have increased the release of toxic metals into the environment. Describe the major ways through which such metals enter the environment and examine their harmful effects on human health.
4. With reference to discarded electronic gadgets and appliances, discuss where such waste originates, what materials it contains, and how it can be recycled or recovered through different methods.
5. Groundwater resources are under increasing pressure in many regions. Analyse the main reasons for excessive withdrawal of groundwater in cities and agricultural areas, and explain the environmental and social consequences.
6. Electronic waste contains a wide variety of materials. Describe the major components commonly present in e-waste and discuss their adverse impacts on human health and the environment when disposal is not managed properly.
7. Climate change is closely linked with extreme weather events. Discuss how global warming influences events such as floods, droughts and heatwaves, and comment on their broader socio-economic implications.
8. Explain how the concept of a circular economy can contribute to sustainable waste management. Describe how product design and responsible manufacturing can help reduce environmental damage.

9. Many pollutants do not degrade easily and persist in the environment for long periods. Describe the ecological and health implications of such pollutants, giving suitable examples.
10. Groundwater is an important source of drinking and irrigation water. Discuss the relationship between climate variability, agricultural practices and long-term changes in groundwater availability.